

Instrument Cataloguer

1 Team Members

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2 Task Overview

We are creating a program to catalogue various orchestral instruments based on their waveform. Many musicians find it very difficult to catalogue instruments because many instruments do not have physical identifiers on them. This task is challenging because the raw waveform must be converted with a Fourier transform into time–frequency features such as a spectrogram.

3 Task Definition

The data we are working with is an audio .mp3 file. This is a single-label, multi-class classification task with 19 classes, one per instrument in the dataset: flute, oboe, cor anglais (English horn), clarinet, bass clarinet, bassoon, contrabassoon, saxophone, French horn, trombone, trumpet, tuba, banjo, cello, double bass, guitar, mandolin, viola, violin. Each clip is one note by one instrument. We have multiple features: fundamental frequencies, timbre, resonance, harmonics, noisiness/dynamics.

4 Problem Description and Impact

Musicians, producers, and archivists work with large collections of recorded audio that are often poorly labelled. Sample libraries can contain thousands of unnamed or inconsistently named clips, and searching recordings by instrument usually depends on someone having tagged them by hand. A model that identifies the instrument from the sound alone would allow these collections to be tagged and searched automatically, and it is a first step towards recognizing instruments within full pieces of music.

This task is challenging for several reasons:

1. **Pitch as a shortcut:** the same instrument sounds very different across its range, and low instruments such as the tuba, contrabassoon, and double bass can be partly identified by their register alone. The model must learn timbre, not just pitch or loudness.
2. **Similar timbres:** instruments in the same family (e.g. viola and violin, oboe and cor anglais) sound very alike.
3. **Class imbalance:** the number of clips per instrument ranges from 74 (banjo) to 1,502 (violin).
4. **Misleading evaluation:** the dataset contains many near-identical recordings of the same note at different dynamics and durations, so a random train/test split overstates performance. Prior work on this dataset reports 97% accuracy using such a split (Mahanta et al., 2021)-.
5. **Clean recordings:** the samples are recorded in studio conditions, so a model trained on them may not transfer to noisier real-world audio.

Our evaluation plan in Section 7 is designed around these challenges: a pitch-only baseline and overlap-

range accuracy for (1), a confusion matrix for (2), macro-F1 and a weighted loss for (3), a grouped split for (4), and a noise robustness test for (5).

5 Data Source(s) and Collection Plan

- Source: <https://philharmonia.co.uk/resources/sound-samples/>
- Metadata: each file's name has the instrument, pitch, note duration, dynamics and articulation.
- We will be using a subset of the full database (brass, strings, woodwinds). This is because percussion instruments do not have that many recordings.

6 Dataset Size and Example Data Points

The collection of instruments has 14,226 clips (one note per file).

Example Data Points

1. **Input:** `flute_A4_1_forte_normal.mp3` (a 1-second flute note A4, forte) **Label:** flute
2. **Input:** `clarinet_C4_1_piano_normal.mp3` (clarinet note, C4, piano) **Label:** clarinet
3. **Input:** `bassoon_G2_1_fortissimo_normal.mp3` (low bassoon note, G2) **Label:** bassoon

7 Proposed Solution

Our pipeline will be as follows: decode the .mp3, convert to mono, resample to 16kHz, trim silences, and normalize clip length by padding or cropping to a fixed length (1-2 seconds, will be determined experimentally).

We can then use the cleaned data to process inputs for a model. We have a baseline where we build hand crafted features like MFCC mean and standard deviation, spectral centroid, rolloff, flatness, zero-crossing rate and attack time for a classical SVM or random forest model. This baseline will be used to compare against a more advanced model using log-mel spectrogram for a small CNN with about 4 conv blocks, which is the main goal we are trying to achieve with the project.

To evaluate the model:

- A grouped split, where we group all clips by '(instrument, pitch)', so the same note at different dynamics or durations never appears in both train and test. Without this the accuracy of the model would be artificially inflated by near identical recordings.
- We use macro F1 score and balanced accuracy as our metrics, since the dataset is imbalanced across instruments. (There are 1,502 violin clips, and while some of the other instruments don't even have clip counts above 100.)
- Tuba, contrabassoon and double bass can be identified partly by register. Therefore, we will evaluate on the pitch range where instruments overlap.

Finally, we will add noise to the test data at increasing levels and plot accuracy against signal-to-noise ratio. The model should be robust to noise, since real-world recordings are not as clean as the dataset.

Libraries: librosa, PyTorch/torchaudio, scikit-learn, panns_inference, audiomentations, pandas/NumPy, matplotlib

Sources of Inspiration

1. J. Engel et al., “Neural Audio Synthesis of Musical Notes with WaveNet Autoencoders” (the NSynth dataset), ICML 2017.
2. A. Eronen and A. Klapuri, “Musical Instrument Recognition Using Cepstral Coefficients and Temporal Features,” Proc. IEEE ICASSP, 2000.
3. Y. Han, J. Kim, and K. Lee, “Deep Convolutional Neural Networks for Predominant Instrument Recognition in Polyphonic Music,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2017.
4. Q. Kong et al., “PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, 2020.
5. D. S. Park et al., “SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition,” Proc. Interspeech, 2019.

Existing Solutions

- Mahanta et al., 2021 (arXiv) The same Philharmonia dataset, 20 classes (percussion merged into one). Fully connected network on MFCCs, 97% accuracy on a random 80/20 split. How we differ: A random split leaks near-duplicate notes into the test set, and there’s no pitch-shortcut or noise testing. So we test robustness directly with added noise.
- Siebert et al., 2016 (PDF) Hand-crafted descriptors with kNN/SVM on Philharmonia (~95%) and on the Iowa samples (much lower, ~84%). How we differ: Shows classic features do well on clean single notes but drop on other recording sets.

8 TA Consultation Summary

We met with our TA, Abdulazeez, during tutorial and walked him through our dataset, the 19 instrument classes, and our plan to compare a classical baseline against a CNN trained on log-mel spectrograms. He felt the project was challenging but appropriate, noting that turning audio into spectrograms makes it an interesting vision-style problem, which confirmed our choice of a CNN as the main model. He advised us to find ways to measure differences between the spectrograms of different instruments. In response, we will compare the average log-mel spectrogram of each instrument and compute distances between classes, which should help explain which instruments the model confuses. He also recommended analyzing results for each class individually, since a large class such as violin could inflate overall accuracy. This is why we report macro-F1 and balanced accuracy alongside per-class results and a confusion matrix, rather than relying on accuracy alone.